

Summary of Findings

Global Literacy Analysis using SQL & Streamlit

Project Objective

The objective of this project was to analyze global literacy trends by combining literacy, illiteracy, GDP per capita, and average years of schooling datasets. The cleaned data was stored in a SQLite database, analyzed using SQL queries, and presented through an interactive Streamlit dashboard with visualizations.

Key Findings

1. Global Literacy Trend

The analysis shows that adult literacy rates have improved over the years in most countries. At the same time, illiteracy rates have steadily declined, reflecting continuous improvements in access to education worldwide.

2. Adult vs Youth Literacy

Youth literacy rates are generally higher than adult literacy rates. This indicates that younger generations have benefited from better educational opportunities and improved literacy programs.

3. Gender Disparity in Literacy

The comparison between male and female youth literacy reveals that the gender gap has reduced in many countries. However, some regions still experience noticeable differences in literacy rates between males and females.

4. Literacy and Economic Development

The analysis indicates a positive relationship between GDP per capita and literacy rates. Countries with stronger economies generally demonstrate higher literacy levels and better educational outcomes.

5. Schooling Years and Literacy

Countries with more average years of schooling tend to achieve higher literacy rates. Although the relationship is generally positive, a few countries show that longer schooling alone does not always guarantee higher literacy, suggesting that education quality and accessibility also play important roles.

6. Country Rankings

The ranking analysis identifies the top-performing and bottom-performing countries based on literacy and illiteracy rates. Developed countries consistently rank among the highest, while several developing countries continue to face educational challenges.

7. Regional Analysis

Regional comparisons show clear differences in literacy levels. Some regions have achieved consistently high literacy rates, while others continue to experience lower literacy and higher illiteracy, highlighting the need for targeted educational initiatives.

8. Illiteracy Analysis

Countries with higher illiteracy rates require greater educational investment and policy support. The analysis also shows that global illiteracy has generally decreased over time.

9. Literacy Projection

Using historical literacy trends, future projections suggest that literacy rates are expected to continue improving in many countries. Countries with already high literacy rates are likely to experience slower but steady growth.

10. Country Profile Dashboard

The Country Profile page provides an interactive view of literacy indicators, GDP per capita, average years of schooling, and youth literacy trends for individual countries. This enables users to compare educational progress over time and better understand country-specific performance.

Conclusion

The project demonstrates that literacy is strongly associated with educational attainment and economic development. Countries with higher GDP per capita and greater average years of schooling generally achieve better literacy outcomes. The Streamlit application provides an interactive platform for exploring global literacy patterns through SQL queries, visualizations, and country-level analysis.

Future Enhancements

- Integrate live data from organizations such as UNESCO or the World Bank.
- Include total population data to estimate the absolute number of illiterate people.
- Apply machine learning models for more advanced literacy forecasting.

- Enhance the Streamlit dashboard with additional filters, comparisons, and downloadable reports.
 - Expand the analysis by incorporating more education-related indicators.
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Tools & Technologies Used

- Python
 - Pandas
 - NumPy
 - SQLite
 - SQL
 - Streamlit
 - Matplotlib
 - Seaborn
 - Plotly
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Outcome

This project successfully transformed raw literacy datasets into an interactive analytical dashboard. It demonstrates practical skills in data cleaning, SQL database management, exploratory data analysis, visualization, and dashboard development, providing valuable insights into global literacy and education trends.